

**ANTEATER POKER
USER MANUAL
VERSION 1.0**

TEAM 21

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GLOSSARY

ANTEATER CARD - NEW CARD ADDED TO THE GAME

BANK - THE AMOUNT OF MONEY A PLAYER HAS

BLIND - AN AMOUNT OF MONEY BET BY A CERTAIN PLAYER. BIG BLIND IS SET TO ONE PLAYER AND SMALL BLIND, WHICH IS HALF A BIG BLIND, IS SET TO THE NEXT PLAYER.

CALL - RAISING YOUR BET TO MATCH THE CURRENT BET

CHECK - MOVE ON WITHOUT ADDING TO THE POT

DEAL - DISTRIBUTE CARDS

DECK - THE CARDS OF THE GAME

FLUSH - 5 CARDS OF THE SAME SUIT

FOLD - QUIT THE ROUND

FOUR OF A KIND - FOUR CARDS OF THE SAME RANK

FULL HOUSE - 3 CARDS OF THE SAME RANK AND TWO CARDS THAT ARE THE SAME OF A DIFFERENT RANK

HAND - CARDS OF THE PLAYER

HIGH CARD - A CARD OF THE MOST VALUE (USUALLY A ROYAL CARD)

PAIRS - TWO OF THE SAME CARDS

POT - THE TOTAL AMOUNT OF MONEY BET BY ALL PLAYERS

RAISE - BET MORE MONEY

RANK - NUMERICAL IMPORTANCE OF EACH CARD (THEIR NUMBER)

ROYAL CARD - ACE, KING, QUEEN, JACK CARD

ROYAL FLUSH - A FLUSH WITH A 10, JACK, QUEEN, KING, AND ACE

SHUFFLE - RANDOMLY CHANGE THE ORDER OF THE CARDS

STRAIGHT - FIVE CONSECUTIVE RANKS IN ORDER

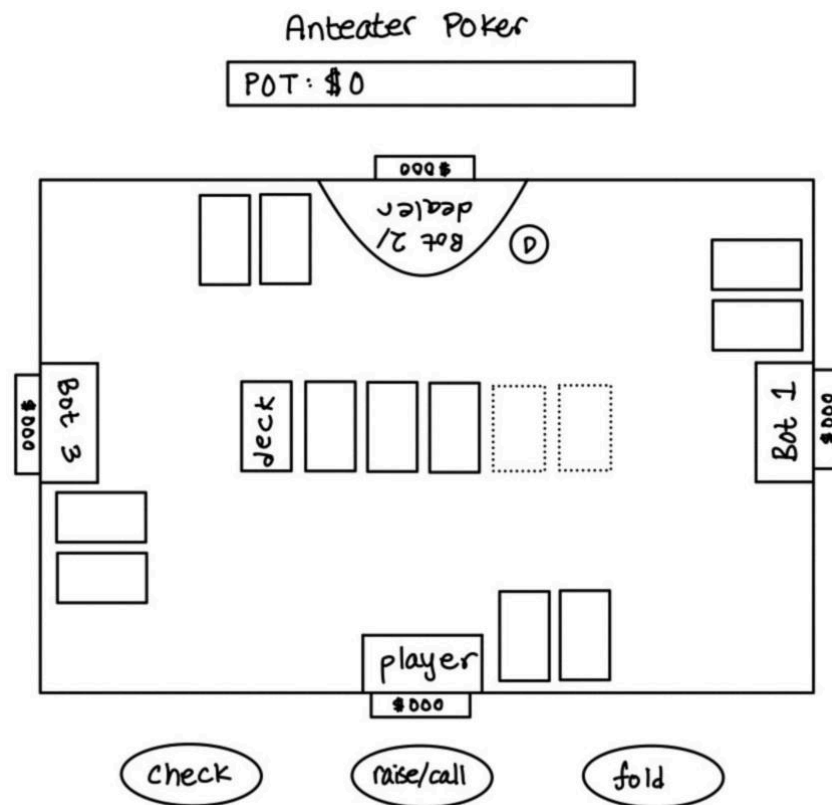
SUIT - HEARTS, DIAMONDS, CLUBS, AND SPADES

THREE OF A KIND - THREE CARDS OF THE SAME RANK

TWO PAIR - A PAIR OF ONE RANK WITH A PAIR OF A DIFFERENT RANK

1.1 Usage Scenario

Anteater Poker is an online variant of Texas Hold'em where friends can connect and play with each other remotely. A session starts with the server being launched on a machine and the players connecting to the server as clients. Each player takes a seat at a virtual table with any remaining seats being filled by bots. Once the players are connected, the server handles the rest of the game by shuffling the deck and dealing cards to each player. Players then take turns checking, raising, calling, or folding as the community cards are dealt. At the end of each round, the pot is distributed accordingly and the deck is shuffled before being redealt.



1.2 Goals

The goal of the program is to simulate an accurate Texas Hold'em game with complete hand evaluation, rules, rotating blinds, and chip tracking. The game should allow for players to connect and play in real-time using simple and intuitive controls (check, raise/call, fold) that make it accessible for players of any experience level. The game should introduce an Anteater card with unique mechanics that make the game more interesting. At the end of each round, it should accurately determine the winner(s) by evaluating each player's best possible hand and distributing the pot properly.

1.3 Features

Anteater Card

- A wildcard unique to Anteater Poker, with 4 copies in the deck.
- When held in a player's hand, it acts as any card of any suit, automatically taking the value most beneficial to that player's hand.
- When it appears on the board, it triggers the Anteater effect: all current community cards are cleared and 5 new ones are dealt, immediately jumping to the final round.

Online Multiplayer

- Players connect to a shared server and compete against each other in real time over a network.
- The server manages all game state including the deck, blinds, betting rounds, and hand evaluation, while clients send and receive updates each turn.

Card Deal

- At the start of each round, two hole cards are removed from the deck and dealt to each player

Blind System

- Small blind and big blind roles are automatically assigned at the start of each round.
- The small blind is set to 50% of the big blind amount.

Betting Actions

- On their turn, players may Check (pass without betting), Raise/Call (increase or match the current bet), or Fold (forfeit the round).
- Betting amounts are deducted from the player's bank and added to the pot in real time.

Progressive Board Reveal

- Community cards are revealed in stages across three rounds, matching standard Texas Hold'em play (flop, turn, river).

Full Hand Detection

- At showdown, the game automatically evaluates each player's best hand from their hole cards and the community cards.
- Detected hands include: high card, pair, two pair, three of a kind, straight, flush, full house, four of a kind, and royal flush.

Stack and Pot Tracking

- Each player's chip stack and the shared pot are tracked and displayed throughout the game, updating after every bet, call, or raise.

Shuffle and Duplicate Prevention

- The deck is shuffled at the start of every game and the program verifies that no card is dealt more than once.

2.1 System Requirements

Before installing and running the program please make sure your system has and meets the following requirements:

- Linux operating system
- GCC compiler
- Make Utility
- Ability to use the Linux terminal

2.2 Setup and Configuration

In order to install the program, please follow the steps below:

Step 1: Extract Files

- a. To extract the files, type the following into your terminal:

```
"gtar xvzf Poker_V1.0_src.tar.gz"
```

Step 2: Change the Directory:

- a. After you have finished extracting the files, change into the program directory. To change into the program directory, type the following into your terminal:

```
"cd poker"
```

Step 3: Compile the program:

- a. Once you have changed to the correct directory, you can now compile the program. To compile the program, type the following into your terminal:

```
"make"
```

Step 4: Run the Program:

- a. The last and final step is to run the program. To run the program, type the following into your terminal:

```
"./poker"
```

2.3 Uninstalling

In order to remove the program from your system, please follow the steps below:

Step 1: Clean the File:

- a. The first step is to clean the compiled files. In order to clean the compiled files, type the following into your terminal:

```
"make clean"
```

Step 2: Remove the Files:

- a. Finally, to remove the files from your system, type the following into your terminal:
"rm-rf poker"

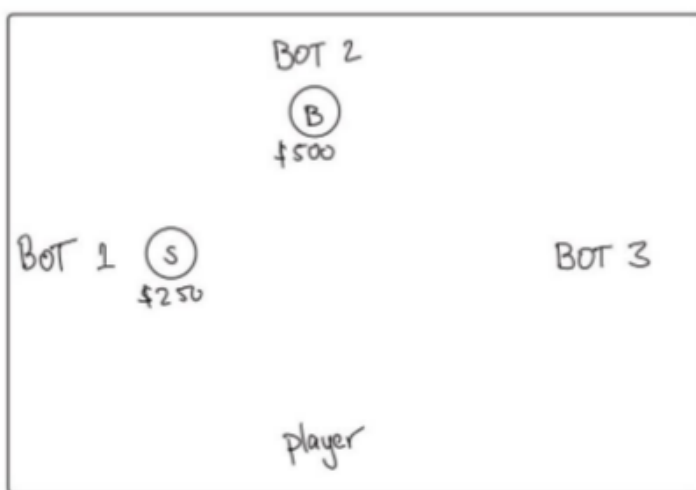
FUNCTIONS

Function 1: Deal

- Deals cards to player
- **Inputs:** player, deck
- **Outputs:** player gets card 1 and card 2
- **Logic:** removes two cards from the deck, and gives it to the player
-

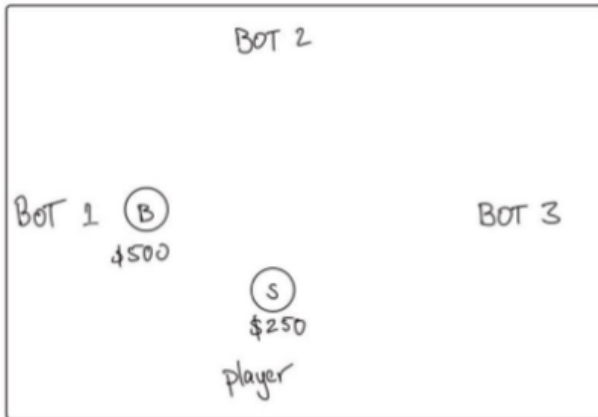
Function 2: set small blind

- Set a player to put in 50% of the big blind
- **Inputs:** player, pot, prev sb
- **Outputs:** money, player++
- **Logic:** Takes away half the amount of big blind from the player and adds it to the pot.
Then switches to the next person

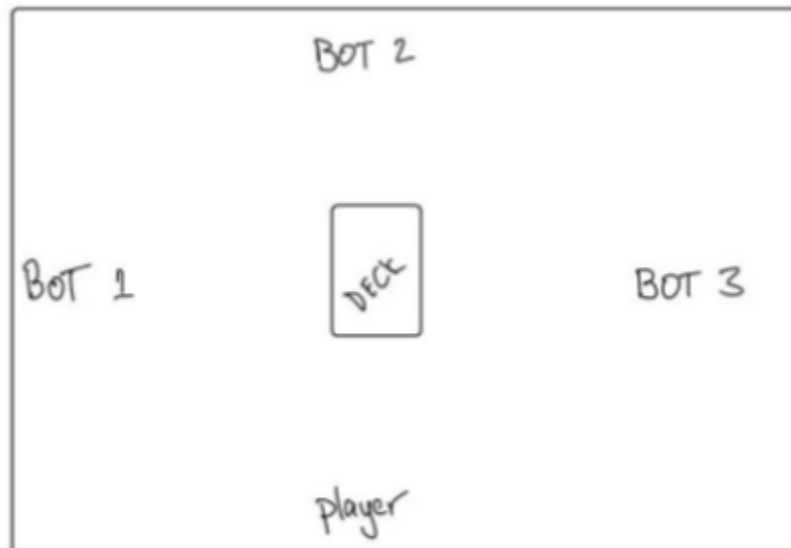


Function 3: set big blind

- Set a player to put in big blind amount
- **Input:** player, pot, prev bb
- **Outputs:** money, player++
- **Logic:** Takes a certain amount away from the player and adds it to the pot. Then switches to the next person

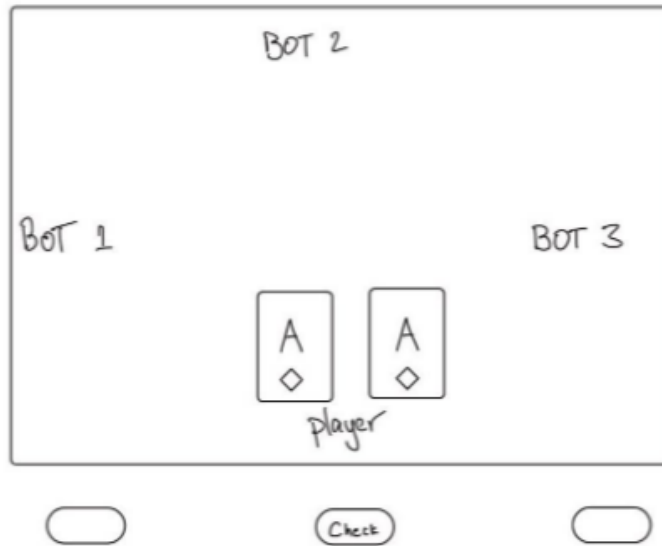
**Function 4: Deck**

- Sets up the deck to be pulled from
- **Input:** player/board, amount of cards needed
- **Outputs:** cards
- **Logic:** have the amount of cards pulled from the deck to a player or board

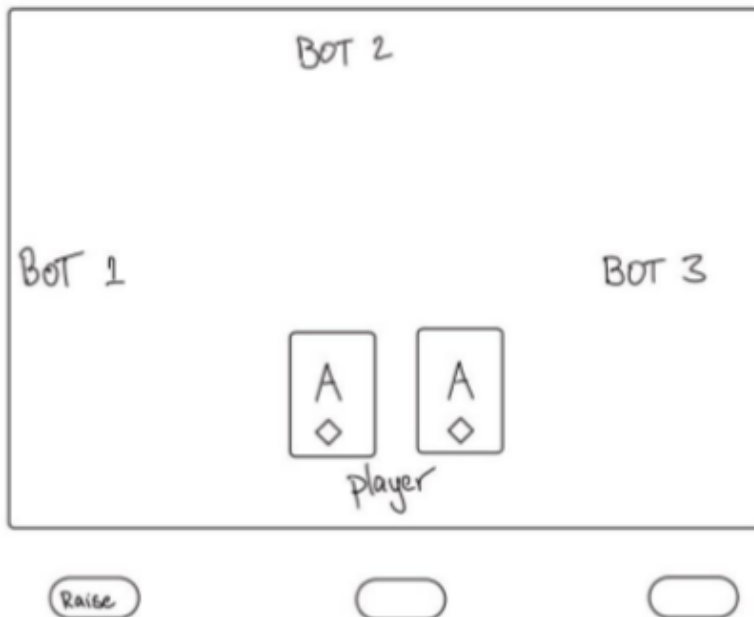


Function 5: Check

- Moves to the next player without placing a bet
- **Input:** bet amount, player, prev player
- **Outputs:** next player
- **Logic:** checks bet amount, moves to the next player if bet amount is equal to the prev player

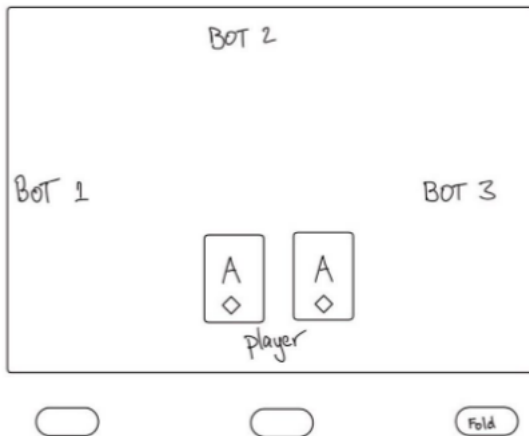
**Function 6: Raise/Call**

- Moves to the next player after placing a bet
- **Input:** bet amount, player, prev player
- **Outputs:** next player, bet amount
- **Logic:** raises the bet amount, removes from the player's bank, moves to the next player



Function 7: Fold

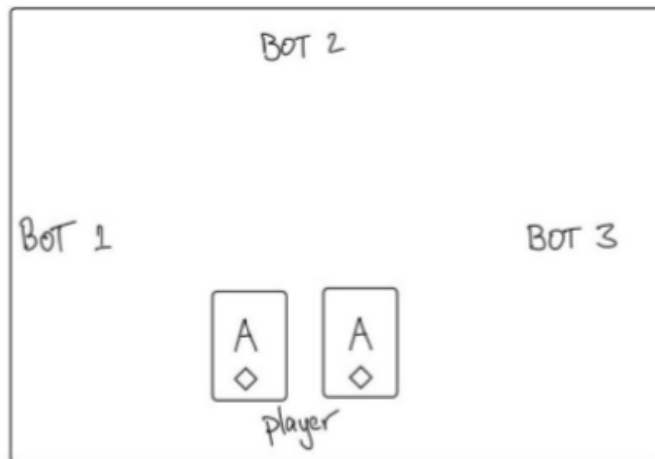
- Makes the player leave the turn
- **Input:** player
- **Outputs:** player is taken out of the round
- **Logic:** makes the player stop playing for the rest of the turn

**Function 8: Shuffle**

- Shuffles the deck
- **Input:** deck
- **Outputs:** deck
- **Logic:** changes the position of each card in the deck

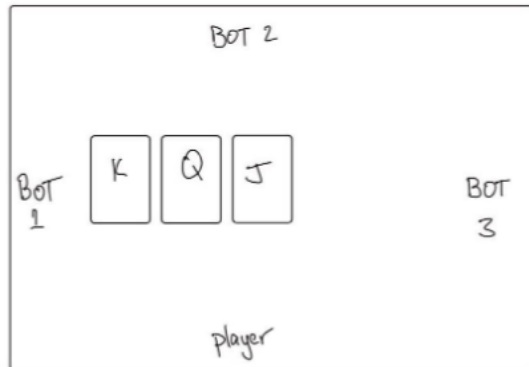
Function 9: hand

- Player's hand
- **Input:** deal
- **Outputs:** cards in hand
- **Logic:** holds the information of the hands in the card

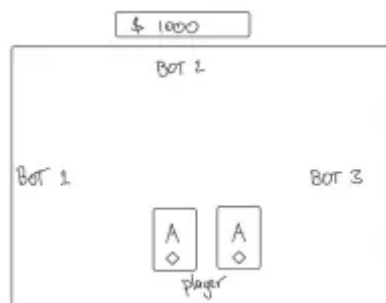


Function 10: board

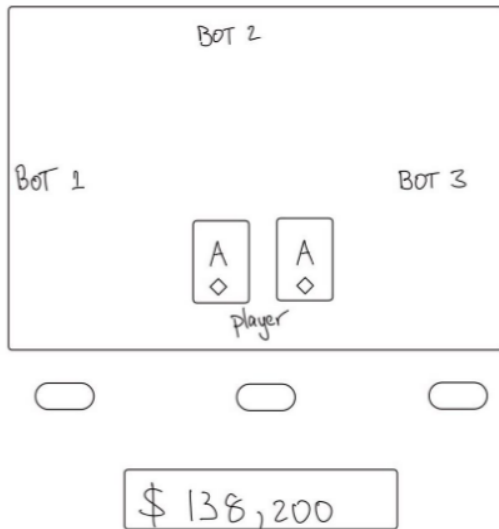
- Display first three cards, then adds the other two
- **Input:** deck, round
- **Outputs:** cards, round
- **Logic:** pulls from deck, displays first three cards, after round 1, pulls from deck and adds card, after round 2 pulls from deck, add 1

**Function 11: Pot**

- Amount of money can be won
- **Input:** small bet, big bet, or raise
- **Outputs:** updated amount of money
- **Logic:** adds money if a bet has been placed or a raise

**Function 12: Bank**

- The amount of money each player has
- **Input:** player
- **Outputs:** player money

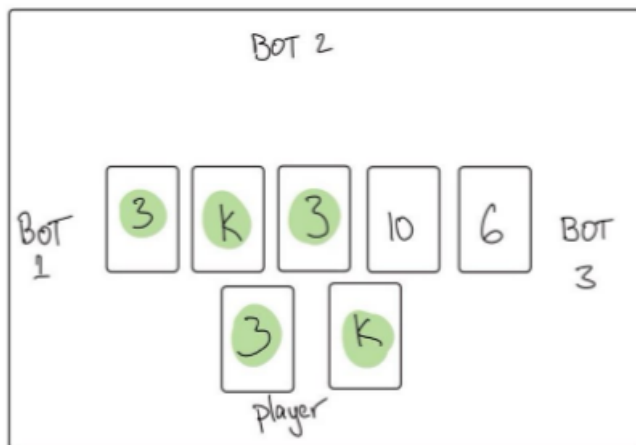


Function 13: Check hand

- Sees who the winner is
- **Input:** hand, board, player
- **Outputs:** value, winValue
- **Logic:** adds the amount of the two cards in hands and sets to value., then adds the board to the player's hand, checks full house, pairs, 3s, 4s, flush, straight, royal flush, adds those values and sets them to winValue

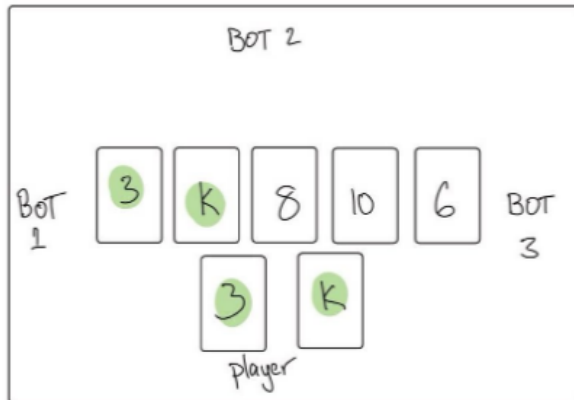
Function 14: Check full house

- Sees if a player has a full house
- **Input:** hand, board, player
- **Outputs:** winValue
- **Logic:** checks to see if a player has 3 cards of the same rank and two cards of a different rank

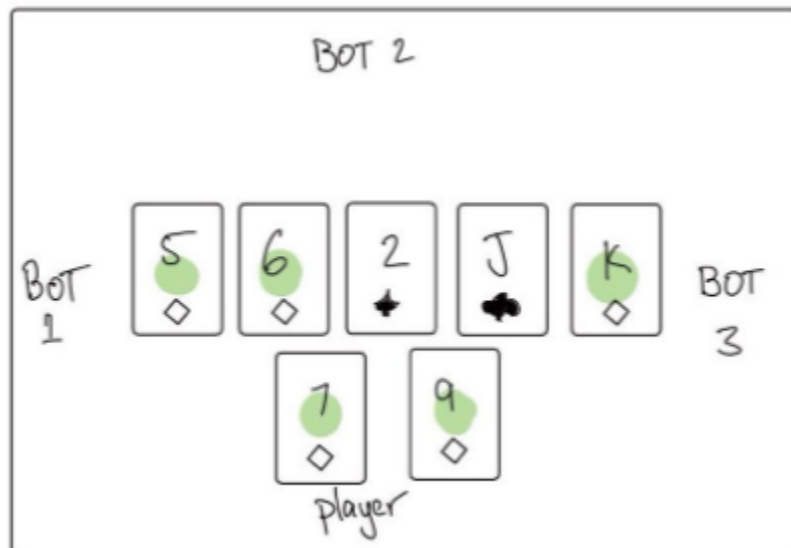


Function 15: Check Pairs, threes, fours, two pairs (could be 4 different functions tbd)

- Sees if a player has a full house
- **Input:** hand, board, player
- **Outputs:** winValue
- **Logic:** checks to see if player has two, three, or four of the same card

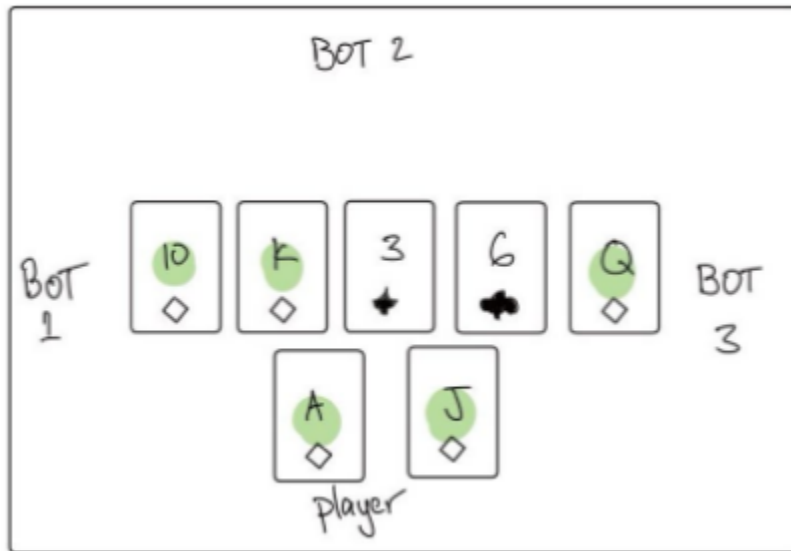
**Function 16: Check Flush**

- See if a player has a flush
- **Input:** hand, board, player
- **Outputs:** winValue
- **Logic:** checks to see if a player has 5 of the same suits

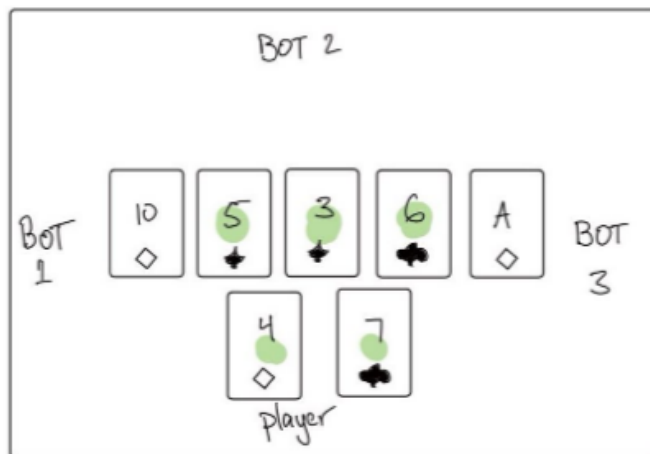


Function 17: Check Royal Flush

- See if a player has a royal flush
- **Input:** hand, board, player
- **Outputs:** winValue
- **Logic:** checks to see if a player has 5 of the same suits and a 10, jack, queen king, and ace

**Function 18: Check Straight**

- See if a player has a straight
- **Input:** hand, board, player
- **Outputs:** winValue
- **Logic:** checks to see if a player has 5 of the same suits



Function 19: Anteater Card

- A new card to spice up the game
- **Input:** deck
- **Outputs:** deck
- **Logic:** If it is a player's hand, it can act as any card and any suit, if it is on the board, the anteater eats all cards on the board and displays 5 new ones. Jumps to the last round.
There are 4 of these cards in the deck



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Anteater Poker is an original adaptation of the traditional poker game, developed exclusively for educational purposes at the University of California, Irvine. The Anteater card and associated rules are defined by the EECS 22L course specification authored by Prof. Rainer Doemer.

ERROR MESSAGES

REFERENCE OF ERROR MESSAGES THAT MAY BE DISPLAYED DURING PROGRAM RUNTIME

ERROR MESSAGE	CAUSE	RESOLUTION
"Invalid input. Please try again"	The player entered a value or command that is not recognized by the program.	Enter one of the valid options shown on the screen.
"Not enough money to raise"	The player attempted to raise more money than they currently have in their bank.	Prompt the player to enter a smaller raise amount that does not exceed the player's available bank.
"Cannot check. Current bet must be matched"	The player attempts to check when another player has already placed a higher bet.	Prompt the player to either call the current bet, raise, or fold instead
"Invalid raise amount"	The raise amount entered is zero, negative, or lower than the required minimum raise.	Prompt the player to enter a valid raise amount greater than the current bet.
"Connection error. Unable to connect to server"	The client could not establish a connection with the server.	Make sure the server is running and that the correct server address is being used.
"Game cannot start. Not enough players connected"	The game was started before the minimum required number of players joined.	Wait for more players to connect or fill remaining seats with bots.
"Player disconnected"	A client lost connection to the server during the game.	Reconnect the player or continue the game with the remaining players/bots

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